Quick quiz: Lectures 3 and 4

Scientific Computing 2026 — 5 minutes — 5 points

Name: _____ Date: _____

Choose **one answer per question**. Each question is worth one point. No calculator is needed.

1. Conditional probability

The joint distribution of X and Y is given below. Find $P(Y = 1 \mid X = 1)$.

	$Y = 0$	$Y = 1$
$X = 0$	0.3	0.2
$X = 1$	0.1	0.4

A. 0.4 **B.** 0.5 **C.** 0.8 **D.** 0.6

2. Correlation and independence

Suppose $\mathrm{Cov}(X, Y) = 0$. Does independence follow (i) in general, and (ii) when X, Y are jointly Gaussian?

A. Yes in both cases.

B. No in general; yes for jointly Gaussian variables.

C. No in either case.

D. Yes in general; no for jointly Gaussian variables.

3. How much does averaging help?

Independent, identically distributed observations have standard deviation 6. What is the standard deviation of their mean for $n = 100$, and what *total* sample size would halve it?

A. 0.6; $n = 200$.

B. 0.06; $n = 400$.

C. 6; $n = 200$.

D. 0.6; $n = 400$.

4. Chebyshev's inequality

A random variable Z has mean 10 and variance 4. What does Chebyshev's inequality tell us about $P(|Z - 10| \geq 4)$?

A. It is at most $1/4$.

B. It is exactly $1/4$.

C. It is at most $1/2$.

D. It is at least $1/4$.

5. What becomes approximately Gaussian?

Let $X_1, \ldots, X_n$ be i.i.d. exponential variables with mean μ and standard deviation σ. Which statement describes the central limit theorem?

A. Each individual X_i becomes approximately Gaussian as n increases.

B. $\bar{X}_n$ converges in probability to μ.

C. $\sqrt{n}(\bar{X}_n - \mu)/\sigma$ converges in distribution to $\mathcal{N}(0, 1)$.